

Notes: Seru, Shumway, and Stoffman (RFS, 2010)

Learning by Trading

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1 Big picture

- **Question.** Do individual investors learn by trading? If they do, do they improve their trading skill, or do poor traders simply learn that they are bad and stop trading?
- **Setting.** The paper uses Finnish Central Securities Depository data covering virtually all Finnish stock trades from 1995 to 2003.
- **Main distinction.** The authors separate two learning channels: **learning by doing** and **learning about ability**.
- **Learning by doing.** Active investors may become better traders with experience, improving returns or reducing biases such as the disposition effect.
- **Learning about ability.** Investors may discover that they are poor traders and stop actively trading; the surviving pool then looks better even if individual ability does not improve much.
- **Key methodological contribution.** The paper tracks individual-level performance and individual-level disposition coefficients over time, then adds investor fixed effects and survival corrections.
- **Main finding.** Simple regressions suggest investors learn quickly, but much of that apparent learning is driven by attrition of low-ability investors.
- **Performance evidence.** Average individual-investor returns are poor, but performance is persistent across investors.
- **Disposition evidence.** Investors are much more likely to sell winners than losers, and investors with stronger disposition effects earn worse subsequent returns.
- **Attrition evidence.** Investors with worse past performance are more likely to exit. Once attrition is accounted for, the estimated effect of experience on performance becomes much smaller.
- **Heterogeneity evidence.** Less sophisticated investors show more improvement, consistent with inexperienced investors learning from trading.
- **Economic takeaway.** Trading teaches investors, but much of the aggregate improvement in the trading population comes from selection: poor traders exit.

2 Data and measurement

2.1 Finnish individual-investor records

- The data come from the central register of shareholdings in Finnish stocks maintained by the Nordic Central Securities Depository.
- Finland has a direct holding system, so the data reflect the official record of stock ownership and trades.
- The sample covers January 1995 through December 2003.
- The data include nearly 1.3 million individual and institutional accounts and more than 22 million trades by individual investors.
- The main analysis focuses on individual accounts that start trading during the sample period.

Interpretation: The data are unusually strong for household finance because they observe actual positions and transactions for a large national market, not only brokerage-house clients.

2.2 Performance measurement

For each purchase, the paper computes the return over the next 30 trading days, truncating the window if the investor sells earlier. The annual performance measure is the average of these purchase-level returns:

$$\bar{R}_{i,t+1} = \frac{1}{N_{i,t}} \sum_{j=1}^{N_{i,t}} R_{i,j,t+1}^{30\text{-day}}. \quad (1)$$

- The main return horizon is 30 trading days because the median holding period is 39 trading days.
- The paper also checks 10-, 45-, and 60-day horizons.
- Returns are sometimes market-adjusted or risk-adjusted with factor models.

Interpretation: The measure asks whether stocks bought by an investor subsequently perform well. It is a trading-skill measure, not a portfolio-level wealth-growth measure.

2.3 Disposition-effect measurement

For each investor-year, the disposition effect is estimated from a hazard model:

$$h_{i,j}(t) = \phi_i(t) \exp(\beta_i^d \text{PriceDummy}_t + \text{controls}). \quad (2)$$

- $h_{i,j}(t)$ is the probability that investor i sells position j on date t , conditional on not selling earlier.
- $\text{PriceDummy}_t = 1$ if the current price is above investor i 's purchase price.
- β_i^d is the investor's disposition coefficient.
- The coefficient is estimated only for investor-years with enough round-trip trades.

Interpretation: A positive β_i^d means the investor is more likely to sell winners than losers. This is the canonical disposition effect.

2.4 Experience measures

The paper uses two main experience measures:

- **YearsTraded:** number of years since the investor first traded.
- **CumulTrades:** cumulative number of trades placed before the current year.

Interpretation: Years traded captures passive experience from observing markets, while cumulative trades captures active learning from placing trades.

2.5 Investor sophistication and controls

- Number of distinct securities traded proxies for diversification and sophistication.
- Trading options proxies for financial sophistication.
- Wealth is proxied by average daily marked-to-market portfolio value.
- Gender and age are observed.

Interpretation: These variables help distinguish learning from changes in the composition of who remains in the sample.

3 Models and empirical specifications

3.1 Simple learning model

The simple learning regression is:

$$y_{i,t+1} = \alpha + \beta_1 Experience_{i,t} + \beta_2 Experience_{i,t}^2 + \delta X_{i,t} + \varepsilon_{i,t+1}, \quad (3)$$

where $y_{i,t+1}$ is either the investor's next-year performance $\bar{R}_{i,t+1}$ or disposition coefficient $\beta_{i,t+1}^d$.

Interpretation: If y is performance, learning predicts positive experience effects. If y is disposition, learning predicts negative experience effects because investors should learn to avoid selling winners too quickly.

3.2 Heterogeneity-adjusted learning model

To control for time-invariant investor skill, the paper estimates versions with individual fixed effects:

$$y_{i,t+1} = \alpha_i + \beta_1 Experience_{i,t} + \beta_2 Experience_{i,t}^2 + \delta X_{i,t} + \varepsilon_{i,t+1}. \quad (4)$$

Interpretation: Fixed effects ask whether the same investor improves as experience accumulates. This separates true within-investor learning from persistent differences across investors.

3.3 Exit and survival model

The survival equation is:

$$s_{it} = I(\alpha_i + \beta_1 Experience_{i,t} + \beta_2 Experience_{i,t}^2 + \delta X_{i,t} + \nu_{i,t} > 0), \quad (5)$$

where $s_{it} = 1$ if investor i continues to trade and $s_{it} = 0$ in the final trading year.

Interpretation: This equation captures learning about ability. If bad traders exit, regressions on surviving traders overstate learning by doing.

3.4 Selection-corrected learning regression

The paper adds inverse Mills ratios from the survival equation to the learning model:

$$\Delta y_{i,t+1} = \beta \Delta X_{i,t} + \rho_{96} I(t = 96) \lambda_{96} + \dots + \rho_{02} I(t = 02) \lambda_{02} + \epsilon_{i,t+1}. \quad (6)$$

Interpretation: Selection correction controls for the endogenous decision to remain an active trader. If the experience coefficient weakens after this correction, attrition was driving the apparent learning.

3.5 Ex post return test for the disposition effect

The Odean-style test compares stocks sold for gains with stocks that could have been sold for losses but were not. A simplified version is:

$$ExPostReturn = \alpha + \gamma_1 SoldGain + \gamma_2 HighDisposition + \gamma_3 (SoldGain \times HighDisposition) + YearFE + u. \quad (7)$$

Interpretation: The interaction γ_3 measures whether high-disposition investors sell winners that subsequently outperform the loser stocks they keep. A positive coefficient means the disposition effect is costly.

4 Identification and empirical design

4.1 What identifies learning by doing

- The same investor is followed over time.
- Experience increases mechanically as time passes and trades accumulate.
- Individual fixed effects remove permanent skill differences across investors.
- The key question is whether performance improves or disposition declines within investor.

Interpretation: Learning by doing is a within-investor phenomenon. It should remain after fixed effects and survival corrections.

4.2 What identifies learning about ability

- Investors who perform poorly are more likely to stop trading.
- If low-ability traders exit, the remaining pool improves even without much within-person improvement.
- The survival model links exit to prior performance, wealth, activity, and experience.

Interpretation: Learning about ability is visible through attrition. It is not enough to show that experienced traders look better; one must ask who left the sample.

4.3 Why the disposition effect is central

- It is a robust behavioral bias.
- It can be estimated separately for investor-years.
- It is costly in the data because high-disposition investors earn worse subsequent returns.
- It provides a behavioral outcome, not only a performance outcome.

Interpretation: If investors learn, they should not only earn better returns; they should also make fewer systematic behavioral mistakes.

4.4 Limits of the design

- The paper cannot observe investors switching to passive index funds after exit.
- The performance measure is purchase-based and does not capture all household wealth dynamics.
- Disposition coefficients require many trades, so they are available only for active investors.
- Experience is not randomly assigned.

Interpretation: The paper's strength is not randomization. Its strength is detailed panel data that allow fixed effects and survival correction.

5 Tables and figures

Visuals are directly cropped from the paper and grouped compactly.

5.1 Table 1 + Figure 1: data and participation

Read:

- Table 1 shows 322,454 new accounts in the full sample and 11,979 accounts with disposition estimates.
- The full sample median account trades for one year and makes three trades.
- Accounts with disposition estimates are much more active: average 222.3 trades and average portfolio value of EUR 58,828.
- The median disposition coefficient is 1.04, implying a median investor is $e^{1.04} \approx 2.8$ times more likely to sell a winner than a loser.
- Figure 1 shows participation rising from about 54,196 accounts in 1995 to 311,013 in 2000, then falling.

Economic message: The paper’s core sample is large, but the investor-years used for disposition estimates are a more active and more sophisticated subset.

Table 1
Summary statistics

	Mean	25th Pctl	Median	75th Pctl	
Panel A: Entire sample (322,454 accounts)					
Number of years with trades	1.9	1.0	1.0	2.0	
Number of securities traded	3.5	1.0	1.0	3.0	
Number of trades	15.4	1.0	3.0	8.0	
Average value of shares traded, EUR	3,447	808	1,653	3,310	
Average portfolio value, EUR	11,588	1,470	2,794	5,856	
Age in 1995	39.3	27.0	39.0	51.0	
Gender (1 = female)	0.39				
Trades options (1 = yes)	0.03				
Panel B: Accounts with disposition estimates (11,979 accounts)					
Number of years with trades	4.4	3.0	4.0	6.0	
Number of securities traded	22.3	12.0	18.0	28.0	
Number of trades	222.3	68.0	117.0	224.0	
Average value of shares traded, EUR	5,356	1,855	3,235	5,759	
Average portfolio value, EUR	58,828	5,102	11,483	26,147	
Age in 1995	35.3	27.0	34.0	44.0	
Gender (1 = female)	0.15				
Trades options (1 = yes)	0.17				
Variable	Mean	t-stat	10th Pctl	Median	90th Pctl
Panel C: Hazard function estimates					
β^d	1.32	6.81	-0.36	1.04	2.57
β^r	-0.04	-0.27	-0.65	0.03	0.93
β^s	0.01	0.46	-0.05	0.00	0.04
β^V	0.25	0.91	-2.62	0.24	3.08

This table presents summary statistics for our data. Panel A includes all individual accounts in our data that started trading during the sample period. Panel B gives results just for those accounts for which we are able to estimate at least one disposition coefficient. We estimate the disposition coefficient only if an individual has placed at least seven round-trip trades in a given year. Number of trades is the total number of trades placed by an investor during the sample period. Average portfolio value is the average marked-to-market value of an investor’s portfolio using daily closing prices. Panel C reports summary statistics for the estimates of the hazard

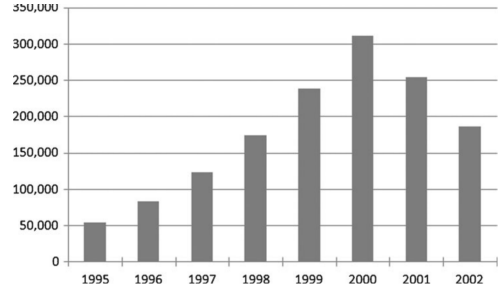


Figure 1
Participation by year
This graph shows the number of accounts that place one or more trades in each year, including both accounts exist at the beginning of the sample and new accounts. There is considerable variation in the number of accounts placing trades over time, from a low of around 54,196 in 1995 to a high of 311,013 in 2000.

(a) Table 1: Summary statistics.

(b) Figure 1: Participation by year.

5.2 Figures 2–4 and Table 2: performance persistence and disposition cost

Read:

- Figure 2 shows performance persistence: investors who perform well early continue to outperform those who perform poorly.
- Figure 3 shows a pronounced kink at zero: investors are more likely to sell positions with positive returns.
- Figure 4 shows that low-disposition investors earn higher subsequent returns than high-disposition investors.
- Table 2 confirms that the disposition effect is costly: the interaction “sold for a gain $\times$ high disposition” is positive at 30-, 45-, and 60-day horizons.

Economic message: The paper first establishes two facts: individual skill persists, and the disposition effect is a costly bias. These facts make learning meaningful.

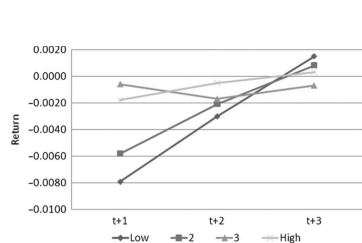


Figure 2
Return persistence
This figure plots the average 30-day returns earned by investors in years following their first purchase. Investors are grouped into quartiles in their first year of trading, and we then calculate average returns for each quartile in subsequent years. Returns are calculated using the approach discussed in the text, and are demeaned by calendar year, which removes the impact of any year fixed effects. Raw returns are reported here, but the resulting risk-adjusted returns are not qualitatively different.

(a) Figure 2: Return persistence.

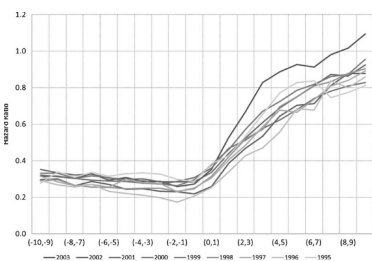


Figure 3
Disposition effect to aggregate
This graph shows how the propensity to sell a stock depends on the stock's return since purchase. Each line plots the regression coefficients from one hazard regression modeling the conditional probability of selling a stock. The coefficients correspond to dummy variables for return bins ranging from $[-10, -9]$ percent to $(9, 10]$ percent.

(b) Figure 3: Aggregate disposition effect.

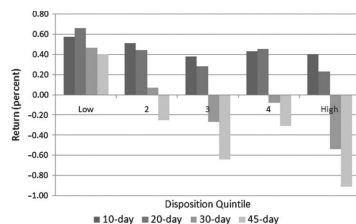


Figure 4
Returns by disposition quintile
This figure shows average 10-, 20-, 30-, and 45-day returns following a purchase for each disposition quintile. Returns are calculated using the approach discussed in the text. Raw returns are reported here, but the results for risk-adjusted returns are not qualitatively different. Returns earned by the lowest quintile (1) are higher than those earned by the highest quintile (5).

(c) Figure 4: Returns by disposition quintile.

Table 2
Ex post return test

	Ex post return horizon		
	30 Days	45 Days	60 Days
Sold for a gain	-0.0164 (0.0021)***	-0.0520 (0.0024)***	-0.0541 (0.0026)***
High disposition	-0.0120 (0.0052)***	-0.0191 (0.0069)***	-0.0145 (0.0084)
Sold for a gain $\times$ high disposition	0.0605 (0.0026)***	0.1063 (0.0033)***	0.1146 (0.0040)***
Year fixed effects	Yes	Yes	Yes

This table shows the results of a regression that effectively implements the same test reported in Odean's (1998) Table 6. The idea is to compare the returns of stocks sold at a gain with those that could have been sold at a loss but were not. In order to see the difference in cost of the disposition effect to investors in the low- and high-disposition groups, we implement a difference-in-difference specification. Using a regression framework instead of simple averages allows us to include year fixed effects to absorb year-specific factors such as average market performance. We define low- and high-disposition groups as the bottom and top quintile of the disposition coefficient. The negative coefficient in the first row indicates that for the low-disposition group the effect documented by Odean is reversed: stocks sold for a gain subsequently underperform. (Robust standard errors, clustered by investor, are presented in parentheses. ***, **, and * denote significance at 1%, 5%, and 10%, respectively.) The difference-in-difference coefficient is significantly positive at the three horizons we consider, indicating that the effect documented by Odean (1998) is stronger for individuals in the high-disposition group.

5.3 Table 3: simple learning and fixed-effect learning

Read:

- In the simple model, experience appears to improve performance and reduce disposition.
- Years traded has a positive coefficient for returns and a negative coefficient for disposition.
- Cumulative trades also predict improvements.
- With investor fixed effects, the estimated learning effects become weaker, especially for returns.

Economic message: Cross-sectional differences in ability explain part of the apparent learning. The same investor does not improve as quickly as simple pooled regressions suggest.

Table 3
Learning models

	Simple learning model		Learning model with heterogeneity			
	$\bar{R}_{i,t+1}$	$\beta'_{i,t+1}$	$\bar{R}_{i,t+1}$	$\beta'_{i,t+1}$	$\bar{R}_{i,t+1}$	$\beta'_{i,t+1}$
YearsTraded _i	0.414 (0.169)***	-0.050 (0.014)***	0.249 (0.401)	-0.010 (0.038)		
YearsTraded _i ²	-0.043 (0.027)	0.007 (0.002)***	-0.022 (0.038)	0.01 (0.003)***		
CumulTrades _i ($\times 10^2$)	0.110 (0.043)***	-0.041 (0.003)***	0.058 (0.059)***	-0.030 (0.005)***		
CumulTrades _i ² ($\times 10^4$)	-0.0005 (0.0003)*	0.0002 (0.00002)***	-0.0009 (0.0004)**	0.0002 (0.0003)***		
NumSec _i	0.096 (0.012)***	-0.011 (0.001)***	0.003 (0.024)	-0.008 (0.002)***		
NumTrades _i	-0.031 (0.003)***	-0.043 (0.007)***	0.005 (0.010)**	-0.029 (0.011)		
PortVal _i ($\times 10^6$)	0.104 (0.061)*	0.099 (0.058)*	-0.006 (0.003)**	0.587 (0.601)		
$\bar{R}_i$		-0.005 (0.001)***	-0.005 (0.001)***	-0.002 (0.001)*		
Individual fixed effects	No	No	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	13,404	13,404	13,404	13,404	13,404	13,404
R^2 (%)	1.6	1.7	1.6	45.7	38.7	38.6

This table presents results for regressions of the form

$$y_{i,t+1} = \alpha_i + \beta_1 \text{Experience}_{i,t} + \beta_2 \text{Experience}_{i,t}^2 + \lambda X_{i,t} + \epsilon_{i,t},$$

where the dependent variable is either the investor's average 30-day return following purchases ($\bar{R}_{i,t+1}$) or the investor's disposition coefficient ($\beta'_{i,t+1}$). In the Simple Learning Model, the α_i are held constant across all investors, while in the model with heterogeneity they are allowed to vary. Experience is measured by either years of experience (YearsTraded) or cumulative number of trades placed (CumulTrades). $X_{i,t}$ is a vector of controls including the number of trades placed by the individual in a given year (NumTrades), the number of securities held by the individual in a given year (NumSec), and the individual's average total daily portfolio value (PortVal). Data are from the period 1995 to 2003. Standard errors are in parentheses, and ***, **, and * denote significance at 1%, 5%, and 10%, respectively.

5.4 Figure 5 + Table 4: attrition and survival controls

Read:

- Figure 5 shows substantial attrition. Many investors exit after one or two years of trading.
- Table 4 estimates survival models and second-stage learning regressions with inverse Mills ratios.
- After survival correction, the YearsTraded effect on performance is no longer statistically significant.
- Inverse Mills ratios are often significant, showing that selection into continued trading matters.

Economic message: Attrition is the paper's main mechanism for learning about ability. Low-performing investors exit, so surviving investors look better.

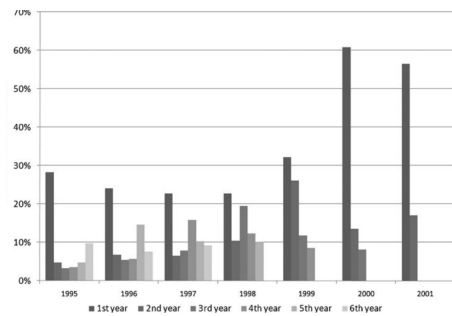


Figure 5
Proportion of accounts who exit
This graph shows the percentage of investors who exit in 1 year, 2 years, ..., or 6 years following the first year in which they place at least seven trades. Exit is defined as placing no further trades during our sample period. Within each year group, the first bar indicates the percentage of investors who exited in the first year after placing a trade, the second bar indicates the percentage of investors who exited in the second year, and so on. Data are missing for later years because we do not know how many investors stopped trading after 2003.

(a) Figure 5: Exit rates.

Table 4
Learning with survival controls

Dependent variable	Selection models (1st Stage)			2nd Stage	2nd Stage
	In-sample: $c_{i,t+1} = 1$			$\Delta \hat{R}_{i,t+1}$	$\Delta \hat{B}_{i,t+1}^d$
YearsTraded _{<i>t</i>}		0.292 (0.199)		0.451 (0.521)	-0.038 (0.042)
YearsTraded _{<i>t</i>} ²		0.028 (0.008)***		-0.009 (0.043)	0.007 (0.003)**
CumulTrades _{<i>t</i>} ($\div 10^2$)		0.341 (0.021)***		0.036 (0.003)***	-0.021 (0.006)***
CumulTrades _{<i>t</i>} ² ($\div 10^4$)		-0.0011 (0.000)***		-0.001 (0.000)	0.002 (0.000)***
$\hat{R}_{t-1}$	0.859 (0.064)***	1.040 (0.082)***		1.050 (0.082)***	-0.211 (0.162)
$\gamma(\hat{R}_{t-1})$		-0.160 (0.062)***		-0.181 (0.061)***	0.047 (0.097)
Trades options		0.257 (0.018)***		0.257 (0.018)***	
Wealthy		0.430 (0.021)***		0.427 (0.022)***	
$I(\text{Inherit} = 1)$				0.134 (0.060)**	
Activeness				0.289 (0.043)***	
<i>Inverse mills ratios:</i>					
γ_{96}				4.868 (0.883)***	-0.128 (0.062)**
γ_{97}				12.57 (1.130)***	-0.141 (0.074)*
γ_{98}				24.03 (1.350)***	-0.388 (0.143)***
γ_{99}				-7.6973 (4.910)	0.291 (0.203)
γ_{00}				4.417 (0.822)***	-0.146 (0.064)**
γ_{01}				7.16 (0.636)***	-0.092 (0.050)*
γ_{02}				12.874 (0.671)***	-0.229 (0.058)***
Observations (<i>N</i>)	36,030	36,030	36,030	11,959	16,188
Time fixed effects	Yes	Yes	Yes		
Other controls	Yes	Yes	Yes	Yes	Yes
Log likelihood	-26077	-22982	-22958		
R ² (%)				12.7	0.83
Joint test of $\rho_i = 0$					
$F(7, N)$				35.80	12.18
$\chi^2 > F$				0.000	0.000

This table reports estimates of selection model regressions with the fixed effects modification developed by Wooldridge (1995). The regressions are of the form

(b) Table 4: Learning with survival controls.

5.5 Figure 6 and Tables 5–7: heterogeneity, intensity, and risk taking

Read:

- Figure 6 shows that trading intensity rises with experience: both number of trades and trade value increase.
- Table 5 shows stronger learning for less sophisticated investors, such as those who do not trade options or have lower wealth.
- Table 6 shows that experienced investors trade more intensively, even after attrition correction.
- Table 7 shows that experienced investors appear to take less factor risk, especially market and size risk.

Economic message: Surviving investors do not simply trade less. They trade more, but in ways that look more experienced and less risky.

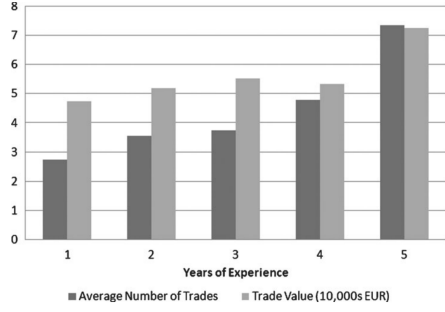


Figure 6
Trading intensity and experience
This figure shows how trading intensity changes with experience. Intensity is measured as the number of trades placed (dark blue) and the total value of trades placed (light blue; in 10,000s of EUR). Results are dem by calendar year to adjust for year fixed effects. The results for the first partial year in which investors trad omitted, so the positive values in the plot are all deviations from that first partial year period. We report re

(a) Figure 6: Trading intensity and experience.

Table 5
Heterogeneity in learning

Classification	Dependent variable							
	$\beta_{t,t+1}^d$				$\bar{R}_{t,t+1}$			
	Mean	Mod. 1	Mod. 2	Obs	Mean	Mod. 1	Mod. 2	Obs
No options trades	1.17	-0.024 (0.003)***	-0.02 (0.001)***	14078	-0.81	0.14 (0.06)***	0.32 (0.13)***	10389
Trades options	0.99	-0.013 (0.004)***	-0.01 (0.004)***	3846	0.1	0.008 (0.04)	0.01 (0.01)	3020
Low wealth	1.14	-0.023 (0.003)***	-0.018 (0.001)***	15426	-1.18	0.05 (0.01)***	0.029 (0.01)***	11411
High wealth	1.11	-0.04 (0.05)	-0.01 (0.012)	2498	1.08	0.02 (0.023)	0.02 (0.021)	1198

This table reports both means and simple learning coefficient estimates from regressions of the form

$$y_{i,t+1} = \alpha + \beta_1 \text{CumulTrades}_{i,t} + \beta_2 \text{CumulTrades}_{i,t}^2 + \delta X_{i,t} + \gamma_i + \epsilon_{i,t}.$$

The variable of interest is either the disposition coefficient ($\beta_{t,t+1}^d$) or returns ($\bar{R}_{t,t+1}$). Regressions using the simple learning model are labeled Mod. 1, while regressions using the adjustment for attrition are labeled Mod. 2. Attrition-corrected regressions include the inverse Mills ratios described in Table 3. For brevity, we report only β_i coefficients in the table. We classify investors as trades options if they trade in options at any point during our

(b) Table 5: Heterogeneity in learning.

Table 6
Trading intensity and experience

	No attrition correction		With attrition correction	
	Num. trades	Val. trades	Num. trades	Val. trades
YearsTraded _i	10.074 (0.618)***	135257 (14091)***	13.32 (0.749)***	22713 (5626)***
YearsTraded _i ²	-0.462 (0.097)***	-9515 (2123)***	-0.896 (0.085)***	-1216 (557)***
Observations	13,266	13,266	13,266	13,266
Individual fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes

This table reports regression results for trading intensity and experience. In Columns 1 and 3, the dependent variable is the number of trades placed by an investor in a year. In Columns 2 and 4, the dependent variable is the value of shares traded, measured in Euros. Attrition-corrected regressions include the inverse Mills ratios described in Table 3. For brevity, we report only the coefficients on experience variables in this table. All regressions include individual and year fixed effects. Data are from the period 1995 to 2003. Standard errors are

(a) Table 6: Trading intensity and experience.

Table 7
Risk-taking and experience

Coefficient	Dependent variables				
	30-day α	β on RMRF	β on SMB	β on HML	β on UMD
YearsTraded	0.58 (0.41)	-0.034 (.0128)**	-0.016 (.0059)**	0.0544 (.0435)	0.0073 (.0250)
YearsTraded ²	0.02 (0.02)	0.00246 (.0012)*	0.00163 (.0005)***	0.0037 (.0028)	-0.00076 (.0019)
CumulTrades ($\div 10^2$)	0.05 (0.02)***	-0.01 (0.002)***	-0.002 (0.0009)***	-0.002 (0.004)	-0.016 (.015)
CumulTrades ² ($\div 10^4$)	-0.0002 (0.0001)**	0.00005 (0.00001)***	0.00001 (0.000006)**	0.00002 (0.00002)	0.00008 (0.00005)
Dep var mean	0.07	0.663	0.028	-0.086	-0.054
Standard dev	3.30	0.288	0.103	0.431	0.440

This table reports the results of fixed effect selection model estimates of regressions of various performance and risk measures on experience measures. The method of these regressions and their associated first-stage estimates are described in Table 4. The dependent variables include each investor's average 30-day risk-adjusted return (α) and each investor's average beta coefficient on four factors—RMRF, SMB, HML, and UMD. These betas are estimated in the standard way, as described in the text. Each regression includes the control variables and inverse Mills ratios described in Table 4, but only the experience variables coefficients are reported in this table. Standard errors are in parentheses, and ***, **, and * denote significance at 1%, 5%, and 10%, respectively.

(b) Table 7: Risk-taking and experience.

5.6 Table 8: aggregate alternative estimation

Read:

- Table 8 estimates disposition learning at the aggregate level using an alternative hazard specification.
- The yearly β_4 coefficients are mostly negative and often statistically significant.
- The 2003 coefficient is -0.123 and highly significant.
- The alternative method supports the core result that investors become less disposition-prone with experience.

Economic message: Even when the disposition model is estimated using pooled trades rather than individual-year coefficients, the learning pattern remains.

Table 8
Disposition estimates at aggregate level using alternative estimation

Year	1995	1996	1997	1998	1999	2000	2001	2002	2003
β_4	-0.074 (0.039)*	-0.093 (0.062)	-0.096 (0.031)***	-0.149 (0.022)***	-0.031 (0.024)	-0.013 (0.011)	-0.063 (0.011)***	-0.069 (0.012)***	-0.123 (0.012)***
Trades	6.6	14.5	26.6	44.6	99.6	251.9	161.9	115.6	131.8
Censored	16%	19%	19%	19%	17%	16%	17%	18%	20%
Accounts	384	713	1360	2412	4953	11585	7445	5416	6056

This table presents learning estimates from pooled proportional hazards models, using a method similar to that of Feng and Seasholes (2005). Each year we pool the trades of all investors, treating them as if they were just one individual and estimating one learning coefficient for the entire population. The model is

$$h(t) = \phi(t) \exp[\beta_1 I(R_{i,t}(t) > 0) + \beta_2 \text{Exper} + \beta_3 \text{Exper}^2 + \beta_4 \text{Exper} I(R_{i,t}(t) > 0) + \beta_5 \text{Exper}^2 I(R_{i,t}(t) > 0)].$$

where Exper is measured in years since first placing a trade and $I(R_{i,t}(t) > 0)$ is an indicator variable that takes a value of one when a stock has increased in price since its date of purchase. For brevity, only the β_4 coefficient estimates are reported. Estimated β_5 coefficients are all insignificant. We also report the number of trades or observations considered by the model (in thousands of trades), the percentage of observations censored (trades not closed by the end of the year), and the number of accounts contributing observations to the model. Standard errors are in parentheses, and ***, **, and * denote significance at 1%, 5%, and 10%, respectively.

6 Short summary

- The paper studies whether individual investors learn from trading experience.
- It uses Finnish transaction and holdings data from 1995 to 2003.
- Individual investors perform poorly on average, but performance is persistent across investors.
- The disposition effect is strong: the median investor is about 2.8 times more likely to sell winners than losers.
- The disposition effect is costly because high-disposition investors earn worse subsequent returns.
- Simple regressions suggest that investors improve with experience.
- Fixed effects and survival corrections show that much of this improvement is due to attrition.
- Investors with poor performance learn about their low ability and stop trading.
- Surviving investors trade more intensively and take less factor risk.
- Less sophisticated investors show more scope for improvement.
- Ignoring attrition overstates how fast active investors become better traders.

7 Conclusion

7.1 Core conclusion

Seru, Shumway, and Stoffman show that investors learn by trading, but not only through gradual skill improvement. A large part of learning comes from selection: low-ability investors discover that active trading is unprofitable and stop trading.

7.2 Economic intuition

If investors differ in ability and do not initially know their type, trading is an experiment. Good outcomes encourage continued trading and higher intensity. Bad outcomes reveal low ability and induce exit. Therefore, the population of active traders improves over time even if individual traders improve only slowly.

7.3 Takeaway

The paper changes how we should interpret evidence of improving investor performance. Better outcomes among experienced traders do not automatically mean that trading makes everyone better. The improvement may mostly reflect that the worst traders leave.